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Genome-wide identification and characterization of long noncoding RNAs involved in apple fruit development and ripening

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ABSTRACT

Long noncoding RNAs (lncRNAs) have been reported to play a regulatory role during fresh fruit ripening. However, the functions of lncRNAs during apple fruit ripening have received relatively little attention. Here, we performed strand-specific RNA-seq and lncRNA identification at four developmental stages of 'Pinova' apple (*Malus domestica*) fruits. A total of 12,570 differentially expressed mRNAs and 3571 differentially expressed lncRNAs were identified among the four fruit developmental stages. Co-localization and co-expression principles were used to explore lncRNA–mRNA target relationships. Interestingly, most lncRNA targeted mRNAs appeared to participate in various fruit ripening pathways involving the cell wall, photosystems, biosynthesis and metabolic processes. To further explore the potential correlations between lncRNAs and mRNAs, a group of 8239 mRNAs and 3566 lncRNAs were used for cluster and co-expression network analysis, and 15 clusters were obtained. Four different clusters were selected for functional analysis and the construction of co-expression networks. mRNAs differentially expressed in the early ripening stages were mainly involved with photosynthesis, whereas those differentially expressed at later stages were associated with metabolic pathways, consistent with the dynamic accumulation of metabolites in ripening fruits. Based on these analyses, we identified a series of genes that play vital roles in fruit ripening and are regulated by lncRNAs; these included *IAA32, SAUR36*, and *PA2*. qRT-PCR analysis also validated these regulatory relationships. Previous studies have shown that lncRNAs can repress miRNA–mRNA targeting by acting as endogenous target mimics (eTMs). We therefore investigated the involvement of lncRNAs in miRNA targeting of mRNA, and eTMs of lncRNAs involved in the targeting of GAMYB (transcriptional factor in anthocyanin synthesis) by miR159 were identified. These results provide insight for future studies of lncRNA regulation during apple fruit ripening.

1. Introduction

Long noncoding RNAs (lncRNAs) are transcripts with a length of >200 nucleotides (nt) that are known to lack coding potential. Some lncRNAs have similarities to mRNAs, as both have a 5′ cap and a 3′ poly (A) tail. lncRNAs can be divided into different types based on their genomic positions: (i) long intergenic ncRNAs (lincRNAs), which are intergenic and do not overlap with any exonic regions; (ii) antisense lncRNAs, which are transcribed from the antisense DNA strand and may be complementary to other mRNAs; (iii) sense intronic lncRNAs, which are derived from the sense DNA strand and do not overlap with exons, and (iv) sense overlapping lncRNAs, which are derived from the sense DNA strand and overlap with exons. Three conserved multi-subunit RNA polymerases in eukaryotes are involved in lncRNA synthesis. Pol I and Pol III are devoted to lncRNA production, and Pol II is used mainly for mRNA but may also function in lncRNA production. The two specific RNA polymerases IV and V are found in plants and produce lncRNAs that recognize and silence transposons (Wierzbicki et al., 2021). Some lncRNAs lack apparent poly(A) tails (Zhang et al., 2019; Zhu et al., 2015), and poly(A) lncRNAs can affect non-poly(A) forms and function under stress (Yuan et al., 2018).

In some earlier studies, lncRNAs were regarded as transcriptional

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'noise' because of their low expression and lack of visible phenotypes. Recently, many studies have provided compelling evidence for lncRNA functions (Ponjavic et al., 2007; Struhl, 2007). The epigenetic significance of lncRNAs has received extensive attention, and their regulatory roles in animals and plants are being refined. Compared with those of plants, the lncRNAs of animals have been relatively well studied. Molecular function analyses have revealed that lncRNAs are potent *cis* and *trans* regulators, and sequence complementation enables them to influence their target genes (Li et al., 2015). lncRNAs act through multiple modalities by interacting with mRNA, DNA, protein, and miRNA to regulate gene expression at the epigenetic, post-transcriptional, and translational levels (Zhang et al., 2019). They can act in different roles in multiple pathways: (i) acting as miRNA target mimics (Thomson and Dinger, 2016; Zhu and Wang, 2012), (ii) recruiting different histone-modifying enzymes to activate or repress gene transcription (Zhang et al., 2019), (iii) contributing to epigenetic silencing via RNA-directed DNA methylation (RdDM) (Chekanova, 2015), and (iv) interacting with splicing factors to regulate alternative mRNA splicing. Studies of lncRNAs have focused mainly on the identification and functional validation of novel lncRNAs (Kang and Liu, 2015; Zhang et al., 2018) and on their relationships with miRNA, mRNA, siRNA, and circRNA interaction networks (Dou et al., 2016; Yan et al., 2020; Zuo et al., 2020). An in-depth understanding of their mechanisms of action will require further research.

The development of high-throughput sequencing enables the in-depth investigation of lncRNA regulatory mechanisms involved in biological processes. Increasing numbers of studies have identified genome-wide lncRNAs in plants, and thousands of lncRNAs have been found in Arabidopsis, rice, and tomato (Correa et al., 2018; Wang et al., 2020). Recent studies have shown that plant lncRNAs help to regulate flowering time, biological pathways, photoperiod sensitivity, photomorphogenesis in red light, phosphate homeostasis, fruit ripening, and responses to biotic and abiotic stresses (Correa et al., 2018; Shafiq et al., 2016; Swiezewski et al., 2009; Wang et al., 2020). The involvement of lncRNAs in fruit development and ripening has been investigated through genome-wide analysis in tomato, apple, and strawberry. More than 5000 novel lincRNAs were expressed in tomatoes during the trans-color stage (Wang et al., 2018). In apple, 168 unique lncRNAs were identified in different tissues and shown to be involved in processes such as energy production, energy conversion, and carbohydrate metabolism (An et al., 2018). Sixty-eight lncRNAs in strawberry were co-expressed with differentially expressed mRNAs related to anthocyanin metabolism (Lin et al., 2018). The regulatory role of lncRNAs has also been explored during fruit development and ripening. In tomatoes, knocking out lncRNA1459 through CRISPR/Cas9 genome editing repressed ethylene production and lycopene accumulation during tomato fruit ripening (Li et al., 2018). ChIP-seq and RNA deep sequencing were used to identify lncRNA targets of RIN (lncRNA2155), and knocking out lncRNA2155 by CRISPR/Cas9 technology delayed ripening of tomatoes (Yu et al., 2019). The lncRNAs LNC1 and LNC2 were targeted by miR156a and miR828a as eTMs and regulated anthocyanin biosynthesis by reducing SPL9 and activating MYB114 in sea buckthorn (Zhang et al., 2018). Similarly, MLNC3.2 and MLNC4.6 functioned as eTMs for miR156a and promoted the expression of SPL2-like and SPL33 transcription factors; the over-expression of eTM and SPL promoted anthocyanin biosynthesis (Yang et al., 2019). Eight lncRNAs appeared to play important roles in fruit ripening and sugar metabolism in different apple tissues (An et al., 2018).

Apple is one of the most popular fruits worldwide, and its fruit development and ripening processes are accompanied by a series of physiological and molecular changes. Previous findings suggest that lncRNAs are involved in apple fruit development and ripening, and they may have a regulatory role in color development and related pathways (Zhang et al., 2018). However, the regulatory roles of lncRNAs in apple fruit development require further investigation. In this study, we identified differentially expressed lncRNAs and analyzed the potential functions of *cis, trans*, and eTM lncRNAs at four fruit developmental stages. We also investigated ripening-related pathways in which they participate and their targeted mRNAs. Taken together, our results provide new insight into the functions of lncRNAs as regulators of the apple fruit ripening process.

2. Material and methods

2.1. Plant materials and sample collection

Apple cultivars 'Pinova' is the offspring of 'Golden Delicious which is the cultivar of the reference genome that may give better results in the bioinformatics analyses, so we used pinova as plant martial for this study. 'Pinova' were cultivated at Northwest A&F University, Yangling, located in the loess plateau of China (34°20′N, 108°24′E). The blooming day is on April 15, which setting to 0 days after anthesis (DAA). We picked apple fruits in different developmental stages: PS1 (27 DAA) at the cell division stage; PS2 (84 DAA) just undergoing starch degradation; PS3 (136 DAA) before fruit coloration; PS4 (165 DAA) when the fruit was fully colored. Four developmental stages of 12 samples were obtained on 12 May (27 DAA), 8 July (84 DAA), 29 August (136 DAA) and 27 September (165 DAA), respectively.

Twelve samples were collected from three healthy 'Pinova' adult trees. Six fruits from different locations' pieces included the pulp and peel were mixed into a single replicate. Likewise, we took 3 replicates from each of the 3 trees and immediately flash froze the samples in liquid nitrogen before storing them at −80°C for subsequent sequencing and RNA extraction (Xu et al., 2020).

2.2. RNA library preparation, reads mapping, expression quantification and differential expression analysis

Twelve samples for following whole transcriptome analysis. Analysis of RNA degradation and contamination by 1% agarose gel. The purity, concentration and integrity of total RNA were assessed using the NanoPhotometer® spectrophotometer (IMPLEN, CA, USA), Qubit® RNA Assay Kit in Qubit® 2.0 Flurometer (Life Technologies, CA, USA) and RNA Nano 6000 Assay Kit of the Bioanalyzer 2100 system (Agilent Technologies, CA, USA).

RNA sample preparations used total of 3 μg RNA per sample as input material. Firstly, Epicentre Ribo-zero™ rRNA Removal Kit (Epicentre, USA) was used for removing ribosomal RNA. Then, using the rRNA-depleted RNA to create twelve sequencing libraries by NEBNext® Ultra™ Directional RNA Library Prep Kit for Illumina® (NEB, USA). For small RNA sequencing, the same RNA per sample used as input material and libraries were generated by NEBNext® Multiplex Small RNA Library Prep Set for Illumina® (NEB, USA).

Following the instructions of the manufacturer, the clustered index-coded samples on the cBot Cluster Generation System were accessed by TruSeq PE Clustering Suite v3-cBot-HS (Illumia). After that, Illumina Hiseq 4000 platform was used for libraries sequencing and generating 150 bp paired-end reads.

The main prediction pipeline is transcript assembly, lncRNAs extraction, protein-coding capacity prediction, and filtering (Fig. S1). After quality control and screening, the clean reads were mapped to the reference genome by HISAT2 v2.0.4(Daccord et al., 2017; Langmead and Salzberg, 2012). The quantification of expression level used Cuffdiff (v2.1.1) to calculate FPKMs of both lncRNAs and coding genes in each sample(Trapnell et al., 2010). FPKM means fragments per kilobase of exon model per Million mapped fragments used following formula to calculate: FPKM = read counts / (mapped reads (Millions) * exon length (KB)). The expression levels of miRNAs were calculated by TPM (transcripts per kilobase of exon model per Million mapped reads) through the following formula(Zhou et al., 2010): Normalization formula: Normalized expression = mapped read counts/total reads*10^6.

Differential expression analysis of 12 samples expression matrix using DESeq2 package to get the significant differences. The significance level was determined by using the corrected p-value (padj). The absolute value of log2(fold change) >1 with padj < 0.05 was defined as the different expression (DE) transcriptions (Table S1). ALL combinations of differentially expressed LncRNAs and mRNAs selected as PS2vs1 (PS2 vs PS1), PS3vs2(PS3 vs PS2), PS4vs3(PS4 vs PS3), PS4vs1(PS4 vs PS1), PS3vs1(PS3 vs PS1), PS4vs2(PS4 vs PS2) (Table S2).

2.3. The prediction of lncRNAs and targeted mRNAs

Firstly, stringTie (1.3.1) was used to assemble the lncRNAs transcriptions (Pertea et al., 2016). The assembled transcripts were filtered by removing the transcripts with uncertain strand orientation and transcripts up to 200 nt in length by Cuffmerge (cufflinks v2.2.1). Then, using Cuffcompare to match with known databases, and filter out the known transcripts. Finally, the novel transcripts were screened by coding potential prediction. Three computational approaches were used to predict the coding potential prediction, respectively, CNCI (Coding--Non-Coding-Index) (v2), CPC (Coding Potential Calculator) (0.9-r2) and Pfam Scan (v1.3) (Bateman et al., 2000; Kong et al., 2007; Sun et al., 2013). Transcripts predicted with coding potential by any of the three tools above were filtered out, and candidates without coding potential were selected as novel lncRNAs. LncRNAs are classified according to the long non-coding RNAs guideline of HGNC (The HUGO Gene Nomenclature Committee).

Cis role is target gene which is position-associated. It shows the target coding gene based on the location relationship with lncRNA, screened within 100k. Expression-associated (*trans*) target gene predicted by the correlation with lncRNA expression (Table S3).

2.4. Co-expression network analysis

To show the clusters and co-expression network of lncRNAs and mRNAs, 8239 mRNAs and 3566 lncRNAs were selected as candidates using k-means and Pearson correlation coefficient (PCC) (Table S4). Firstly, all transcripts were divided into 15 clusters by k-means (Table S5). The heatmaps of k-means centers and FPKM were made by TBtools (Chen et al., 2020). And the top 500 connections (less than 500 with all) which PCC$>$0.9 and p.adj$<$0.01 to construct co-expression networks (Table S6). The lncRNAs and the neighboring key genes from the extracted connections were used to construct simple co-expression networks in Cytoscape (Saito et al., 2012). The importance of mRNAs and lncRNAs uses cytoHubba to get the top 30 which red presents the more importance (Chin et al., 2014).

2.5. Analysis of GO and KEGG function enrichment, and pathways related to DE-lncRNAs

Enrichment analysis of differentially expressed genes or lncRNA target genes by Gene Ontology (GO) is implemented by the GOseq and clusterProfiler R package (Young et al., 2010; Yu et al., 2012). GO terms with corrected p-values less than 0.05 were considered as significantly enriched by differentially expressed genes. KOBAS software tests the enrichment of differentially expressed genes or lncRNA target genes in the KEGG pathway (Kanehisa et al., 2008).

Co-expressed DE-mRNAs targeted by DE-lncRNAs at PS2vs1, PS3vs2 and PS4vs3 were selected for analysis in various biological pathways. MapMan annotation were constructed on Mercator4 V3.0 (https://plabipd.de/portal/mercator4) using the reference genome. The DE-mRNAs were submitted to MapMan (3.6.0RC1) for analysis of the biological pathways involved.

2.6. Construction of lncRNA-miRNA-mRNA network

To predict the function of lncRNAs as eTMs, we used Targetfinder and psRNATarget (https://www.zhaolab.org/psRNATarget/) to predict targeted relationships based on the complementary pairs between miRNAs-lncRNAs and miRNAs-mRNAs. All members with potential targeted relationships were used for constructing the network by Cytoscape. Based on the ceRNA theory, lncRNAs that may be eTMs were selected in the network for verification.

2.7. RNA extraction and RT-qPCR analysis

Different four stages of fruit RNA extraction using RNAprep Pure Plant Kit (Polysaccharides&Ployphenolics-rich, DP441) (Tiangen Biotech, Beijing, China). The primers used in the experiment were designed using Primer-BLAST on the NCBI website (https://www.ncbi.nlm.nih.gov/tools/primer-blast/) (Table S7). The first-strand cDNA was synthesized by using HiScript® II 1st Strand cDNA Synthesis Kit (+gDNA wiper) (Vazyme, Nanjing, China). Real time Quantitative PCR (RT-qPCR) analysis was carried out applying ChamQ SYBR qPCR Master Mix (Vazyme, Nanjing, China) on a Bio-Rad CFX96TM system.

3. Results

3.1. Genome-wide identification and characterization of lncRNAs in different apple ripening stages

To explore the possible regulatory role of lncRNAs in apple fruit development and ripening, strand-specific RNA-seq (ssRNA-seq) was used to identify lncRNAs from four fruit developmental stages (PS1, 23 days after anthesis (DAA); PS2, 84 DAA; PS3, 136 DAA; and PS4, 165 DAA). Paired-end ssRNA-seq was performed on three biological replicate samples from each stage, and 1.09×10^9 clean reads from the 12 samples were used to identify lncRNA candidates. A total of 52,919 lncRNAs were identified across the four fruit developmental stages. Sequence length, open reading frame (ORF) length, and exon number of the identified lncRNAs and mRNAs are shown in Fig. 1B–D. The results showed 91.38% of the predicted lncRNAs contained only one exon, with an average of 1.21 exons (Fig. 1B). The mean sequence length of the lncRNAs was 469.42 nt, and most were in the range of 230–240 nt (Fig. 1C). The average ORF length was 77.41 bp, and most ORFs were approximately 70 bp in length (Fig. 1D). These results show that lncRNAs have fewer exons, shorter sequence lengths, and shorter ORFs than mRNAs (Fig. 1B–D).

Circos software was used to visualize the distribution of lncRNAs across the apple chromosomes (Fig. 1E). The inner histograms show that lncRNAs are distributed more uniformly than mRNAs on the apple chromosomes. Most regions with a low density of mRNAs showed no difference in the density of lncRNAs. Several chromosomes, such as 1, 4, 5, and 14, showed similar distributions of mRNAs and lncRNAs in specific regions, and there appears to be some correlation between lncRNAs and mRNAs in these regions. The outer heatmap shows fragments per kilobase per million mapped reads (FPKM) of lncRNAs at different fruit developmental stages. FPKM values differ among developmental stages in some regions, suggesting that lncRNAs in these regions may play a role in apple ripening. Based on the genomic locations of lncRNAs relative to those of protein-coding genes, we classified the lncRNAs into four different types using HGNC (Fig. 1F): 41,737 lincRNAs (lincRNAs, 78.9%), 8764 antisense lncRNAs (16.6%), and 2418 sense overlapping lncRNAs (4.6%). No sense intronic lncRNAs were identified in this analysis. Although most lncRNAs were located in intergenic regions, some were located near coding genes and may have intersected with neighboring mRNAs.

3.2. Genes differentially expressed during apple fruit development

To identify specific lncRNAs that play important roles during fruit ripening, we calculated the FPKM values of all mRNAs and lncRNAs at four fruit developmental stages. We constructed an expression heatmap of the 12,570 mRNAs and 3571 lncRNAs that were differentially

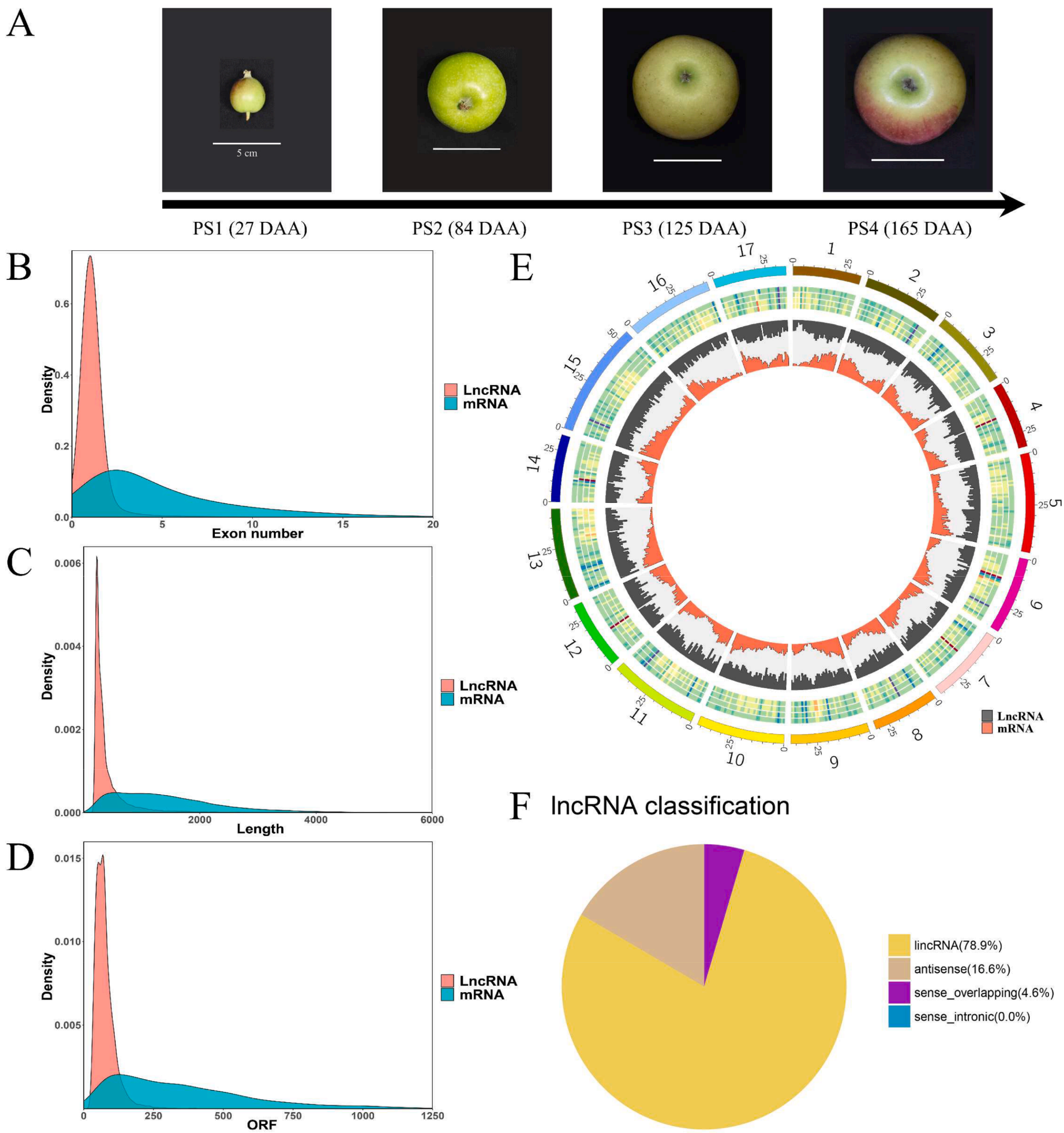

Fig. 1. Characteristics of Malus × domestica lncRNAs. (A) The phenotypes of 'Pinova' apple fruit developmental stages. (B) The density of exon numbers for mRNAs and lncRNAs. (C) Sequence lengths of lncRNAs and mRNAs. (D) ORF lengths of lncRNAs and mRNAs. (E) Circos diagram of lncRNA and mRNA density on apple chromosomes and FPKM values of lncRNAs at four apple fruit developmental stages. The outermost layer represents the apple chromosomes. The second layer is a heatmap of FPKM values for stages PS1, PS2, PS3, and PS4 (outer to inner). The innermost layer is a histogram in which the outer grey bars show the distribution of lncRNAs and the inner red bars show the distribution of mRNAs. (F) Classification of lncRNAs into four types.

expressed among developmental stages (Fig. 2A and B). Then, the differentially expressed mRNAs and lncRNAs were clustered by their expression patterns. More than two-thirds of lncRNAs had higher expression levels at PS1 than at other stages, indicating an overall trend towards declining lncRNA expression levels during fruit development (Fig. 2C). The expression patterns at PS1 and PS2 were clearly distinct from those at PS3 and PS4. PS3 and PS4 showed similar expression patterns, probably because there were fewer dramatic changes at the molecular level at these late fruit developmental stages. This result was also reflected in the Pearson correlation coefficients and the principal component analysis (PCA) (Fig. S3).

Next, DESeq2 was used to identify differentially expressed lncRNAs

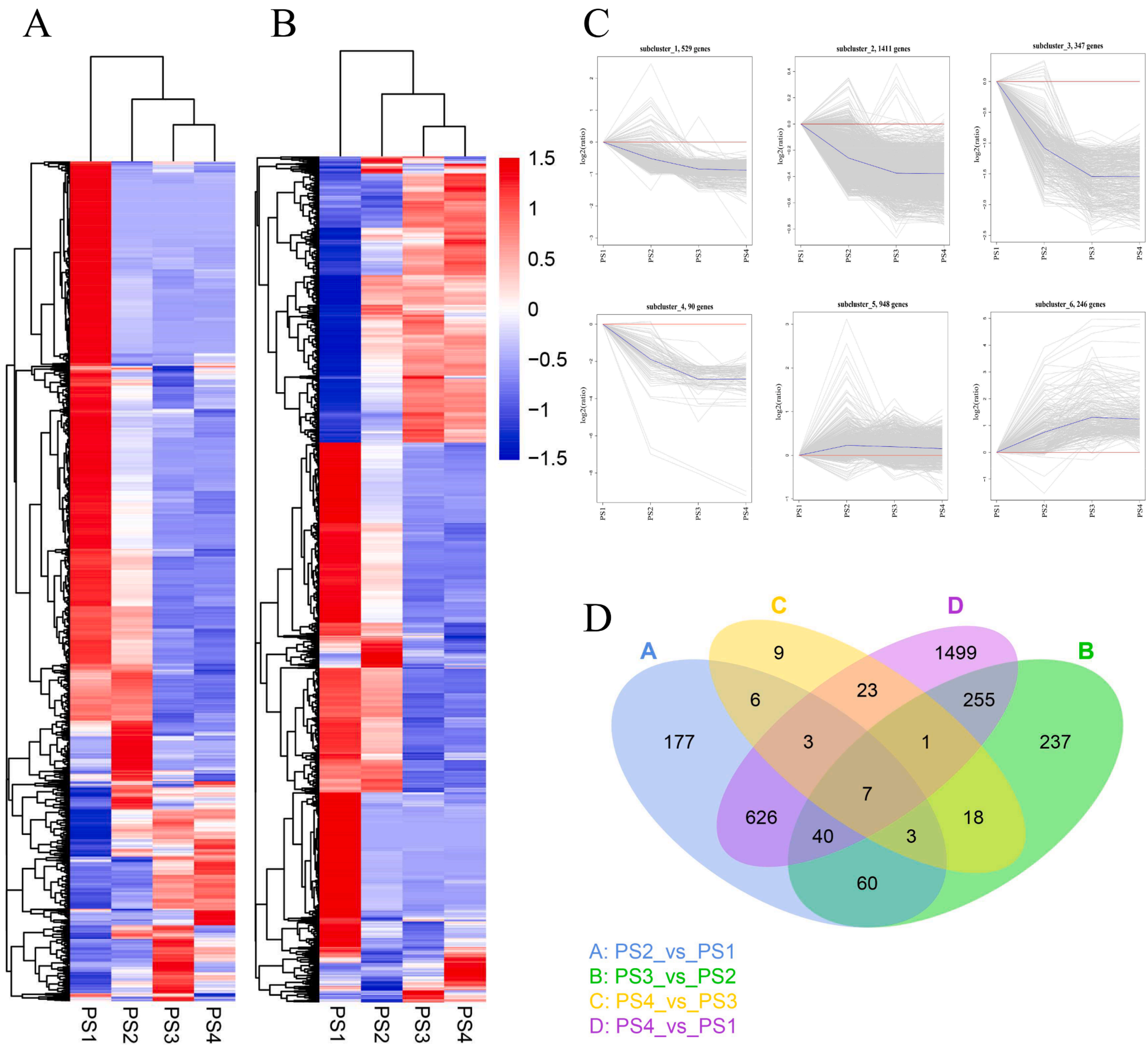

Fig. 2. Differentially expressed mRNAs and lncRNAs. (A) A hierarchically clustered heatmap of differentially expressed lncRNAs. (B) A hierarchically clustered heatmap of differentially expressed mRNAs. (C) Line charts showing the expression patterns of lncRNAs from the heatmap in (A). (D) Venn diagram of differentially expressed lncRNAs from four comparisons among fruit developmental stages.

between specific developmental stages: PS2vs1 (PS2 vs. PS1), PS3vs2 (PS3 vs. PS2), PS4vs3 (PS4 vs. PS3), and PS4vs1 (PS4 vs. PS1) (Fig. S2). Notably, there were more differentially expressed lncRNAs in PS2vs1 than in PS3vs2, similar to the trend in mRNA results. There were fewer upregulated lncRNAs than downregulated lncRNAs in the four comparisons. Differentially expressed lncRNAs that overlapped among the four comparisons were visualized with a Venn diagram and the common lncRNAs among the four comparisons were considered to have roles in fruit development (Fig. 2D).

To investigate the functions of lncRNAs during fruit development, we performed a Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis of differentially expressed mRNAs at the four developmental stages. Documenting the functions of differentially expressed mRNAs at each time point provided insight into the biological processes that occurred at each stage of ripening. Enriched KEGG pathways included biosynthesis of secondary metabolites, fatty acid metabolism, and plant hormone signal transduction (Fig. S4). The combined analysis of lncRNA and mRNA expression is important for revealing the roles of lncRNAs in fruit ripening.

3.3. Functional prediction of lncRNA targets in apple

Previous studies have shown that lncRNAs can regulate protein-coding genes in *cis* and *trans*-acting modes (Kang and Liu, 2015; Kopp and Mendell, 2018; Zhan et al., 2016). Position-associated and expression-associated target gene predictions were used to identify potential lncRNA target genes and explore their functions. For position-associated target gene prediction, we identified candidate *cis* target genes based on the relative locations of lncRNAs and mRNAs within a 100-kbp window. For expression-associated target gene prediction, we identified mRNAs whose expression was correlated with that of lncRNAs. Based on the co-expression results, we assigned GO annotations to the candidate lncRNA target genes. PS2vs1 target genes showed strong enrichment in biological process GO terms associated with biological regulation, the regulation of cellular processes, and signal transduction. PS3vs2 target genes showed enrichment in biosynthetic and metabolic processes, whereas PS4vs3 target genes showed enrichment in protein and cellular components. Enrichment of cellular component GO terms was similar in all three comparisons; enriched GO terms were associated with the photosystems, the nucleus, and related complexes. For molecular function GO terms, PS2vs1 and PS3vs2 target genes showed enrichment for molecular binding functions and also for malic acid, fructose, and carbohydrate phosphatase activity, whereas PS4vs3 target genes mainly showed enrichment in hydrolase and oxidoreductase activity (Fig. 3). These results suggest that lncRNAs differentially expressed during fruit development may be involved in the regulation of metabolism and molecular binding.

To pinpoint lncRNAs that were most likely to have an impact on the four developmental stages, we selected mRNAs that were co-expressed with lncRNAs and that had absolute $\log_2$(fold change) values > 10, and we used them for further functional analysis. PS2vs1 and PS3vs1 target genes were mainly associated with cell wall biogenesis, hormone

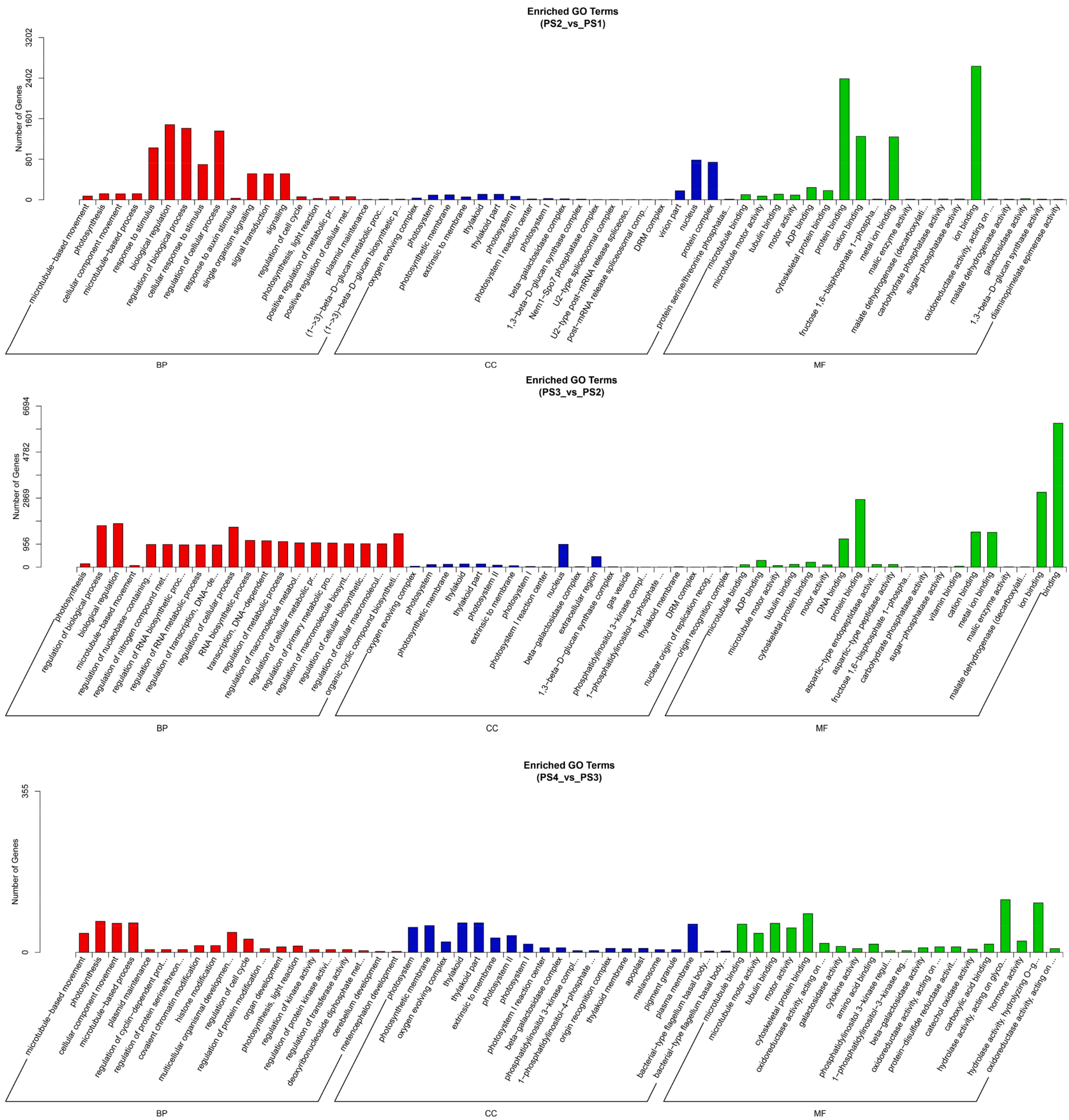

Fig. 3. GO enrichment analysis of target genes co-expressed with lncRNAs. GO enrichment results for target genes co-expressed with differentially expressed lncRNAs in three developmental stage comparisons. Enriched BP (biological process), CC (cellular component), and MF (molecular function) terms are shown.

metabolic processes, and microtubule-based process. PS3vs2 and PS4vs1 target genes were primatily enriched in terms associated with photosynthesis and cell cycle processes. There were fewer differentially expressed genes in the PS4vs3 comparison, and the lncRNA target genes were enriched in RNA modification and regulation of DNA recombination, unlike target genes from the other stage comparisons. PS4vs2 target genes were mainly enriched in GO terms related to polysaccharide biosynthetic process and photosynthesis.

There were fewer enriched GO terms for target genes identified by position-associated prediction than for target genes identified based on co-expression. Nonetheless, there were some notable enriched terms, especially those associated with molecular binding, including phosphate-containing compound metabolic process and transferase activity (Fig. S5). There were no significantly enriched biological process terms in the PS2vs1 comparison, but terms associated with phosphorylation and protein modification processes were enriched in the PS3vs2 and PS4vs3 comparisons. All three comparisons showed enrichment of cellular component terms associated with the cell wall, which is related to fruit firmness. PS2vs1 target genes showed significant enrichment of the molecular function terms catalytic activity and transferase activity.

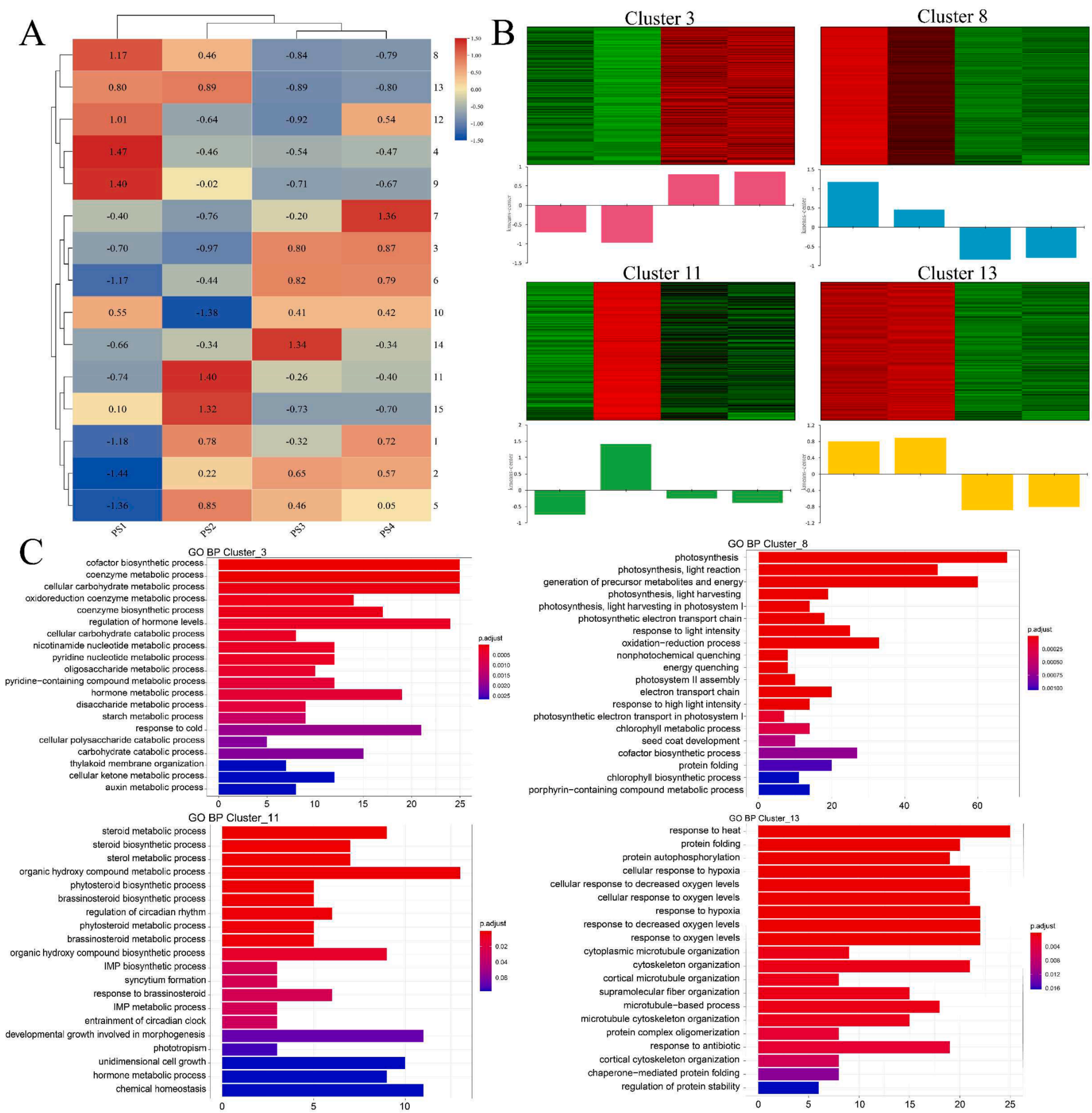

Fig. 4. Visualization of stage-specific lncRNAs and mRNA expression clusters and their functional enrichment. (A) A heatmap of all clusters based on k-means centers. (B) The expression heatmaps and histograms of four clusters with characteristic patterns at different stages of fruit development. (C) Functional enrichment of four clusters with characteristic expression patterns.

In addition to transferase activity, the PS3vs2 target genes also showed enrichment in protein kinase function. In the PS4vs3 comparison, target genes were enriched in kinase and nucleoside binding terms. Taken together, these results indicate that lncRNAs may target genes that participate in cell wall processes and phosphorylated protein modifications, consistent with the GO enrichment results for target genes identified by the co-localization approach.

3.4. Co-expression cluster analysis predicts pathways targeted by lncRNAs

LncRNAs serve a regulatory function by targeting mRNAs, and expression correlations may reveal the relationships between them. We performed k-means clustering to identify potential correlations between lncRNAs and mRNAs. Before the analysis, differentially expressed genes with low expression (FPKM $\leq$ 0.05) in more than one sample from the same stage were removed, and 8239 mRNAs and 3566 lncRNAs were retained for clustering analysis. In total, 11,805 sequences were classified into 15 clusters with different expression patterns in four fruit developmental stages. The number of mRNAs ranged from 237 in cluster 1 to 1803 in cluster 9; the number of lncRNAs ranged from 16 in cluster 10 to 973 in cluster 4 (Fig. 4A). Most were highly expressed in PS1, and their expression gradually decreased in most clusters. Only a small number of transcripts were included in clusters with high expression at PS3 and PS4, consistent with the results of differential expression analysis.

To further investigate the functions of mRNAs and lncRNAs from different clusters, we performed GO enrichment analysis of genes from selected clusters: cluster 13, which showed high expression in PS1 and PS2, cluster 8 (high expression in PS1), cluster 11 (high expression in PS2), and cluster 3 (higher expression in PS3 and PS4) (Fig. 4B). Cluster 3 (731 mRNAs, 124 lncRNAs) was mainly related to carbohydrate metabolic process, starch metabolic process, and coenzyme metabolic process. Cluster 13 (1056 mRNAs, 203 lncRNAs) was enriched in protein folding, response to oxygen levels, and cytoskeleton organization. Cluster 8 (1663 mRNAs, 381 lncRNAs), whose transcripts were highly expressed only during PS1, was related to photosynthesis and energy (Fig. 4C). Clusters 8, 9, and 4 had the largest number of transcripts, indicating that most transcripts active before fruit expansion have fewer roles later in fruit development. Finally, we examined cluster 11, whose transcripts were highly expressed during PS2. Genes in this cluster were related to sterol and steroid metabolic processes (Fig. 4C). In summary, lncRNAs differentially expressed during PS1 and PS2 appeared to be involved in photosynthetic and cell structure–related pathways, and lncRNAs differentially expressed during PS3 and PS4 appeared to be involved in the accumulation of carbohydrates.

3.5. Regulatory networks of key genes targeted by lncRNAs in different clusters

To identify the potential functions and interactions of lncRNAs and mRNAs in different clusters, we exported information on the Pearson correlation coefficients between cluster members (PCC $>$ 0.9, P.adj $<$ 0.01). Because the number of connections varied among different clusters, the top 500 connections of the four clusters were used for key gene calculations. The lncRNAs and their neighboring key genes from the extracted clusters were used to construct simple co-expression networks in Cytoscape.

Among the co-expression networks of the four clusters, the most important key gene (MD15G1353000), which connected with four lncRNAs, encoded a pyridoxal phosphate (PLP)-dependent transferase superfamily protein with an important role in fruit ripening (Kumar et al., 2016). The key genes MD02G1079200 (beta-galactosidase 12), MD10G1328100 (ethylene-forming enzyme), and MD09G1069500 (lipoxygenase 1), which were connected with TCONS_00052251, TCONS_00107551, and TCONS_00022555, were associated with ripening and carbohydrates accumulation in cluster 3 (Fig. 5A). MD05G1311600 (ERF12-like protein) and MD07G1117200 (YABBY family protein) were related to TCONS_00071339 at flowering (Chandler and Werr, 2020) and early fruit development (Zhao et al., 2020) in cluster 8 (Fig. 5B). MD02G1149000 and MD15G1032800 (both of cytochrome P450 protein) were connected to TCONS_00020970 and TCONS_00129163 in cluster 11 and were involved in synthesis of pigments (Fig. 5C). From cluster 13, MD07G1262500 encodes a phosphoenolpyruvate carboxylase-related kinase 1 that was connected with TCONS_00063634 in photosynthesis (Fig. 5D).

3.6. lncRNAs are involved in primary and secondary metabolism during apple fruit development

The results of functional enrichment and network analysis showed that mRNAs targeted by lncRNAs were involved in many processes of primary and secondary metabolism, but their specific regulatory pathways remained unclear. To further delineate the specific pathways regulated by the lncRNAs, we used MapMan to build a model diagram of the targeted differentially expressed mRNAs and differentially expressed lncRNAs. The overall metabolic pathway and secondary metabolic pathway maps were constructed using three comparisons: PS2vs1, PS3vs2, and PS4vs3. The overall metabolic map shows that most of the genes related to the cell wall and photosynthesis were downregulated in PS2vs1 and PS3vs2. By contrast, genes related to carbohydrate metabolism, lipid metabolism, and secondary metabolic pathways were upregulated in these two comparisons, and the number of upregulated genes gradually increased with fruit development. There were fewer differentially expressed genes in PS4vs3, but they showed higher enrichment in pathways related to phenolics (Fig. S6). The fruit ripening process was strongly associated with secondary metabolism. Partial upregulation of terpenoid-related genes was observed in the secondary metabolic pathway during the transition from stage 1 to stage 3 (PS2vs1 and PS3vs2). Moreover, phenolic-related genes were upregulated in all three comparisons. These results were broadly consistent with metabolic variation trends during the stages of apple fruit development (Fig. 6).

3.7. Identification of lncRNAs related to fruit development

To further verify the accuracy of the genomic data, we selected 10-kb co-located and co-expressed high-confidence lncRNA–mRNA pairs for qRT-PCR validation. The remaining co-expressed DE-lncRNAs were studied as *trans* lncRNAs. Three of the ripening-related, differentially expressed lncRNA–mRNA pairs were selected for expression validation: MD13G1035200–TCONS_00131720, MD02G1205700–TCONS_00025525, and MD03G1013600–TCONS_00032426. The proteins encoded by these three high-confidence targeted mRNAs were the auxin-responsive protein IAA32, SAUR-like auxin-responsive protein SAUR36 in the auxin signaling pathway and peroxidase A2-like (PA2) protein related to phenylpropanoid biosynthesis. Three differentially expressed *trans* lncRNAs were selected (TCONS_00130559, TCONS_00104644, TCONS_00076972) because their targeted mRNAs were enriched in primary and secondary metabolic processes. Three highly differentially expressed lncRNAs (TCONS_00022029, TCONS_00116832, TCONS_00139880) were also selected for qRT-PCR validation; their predicted target mRNAs included lipoxygenase, glutaredoxin, and transcription factor ERF118. All three *cis*-regulated genes showed the highest expression at PS1, then gradually decreased in expression as fruit development continued. In all cases, the qRT-PCR results were broadly consistent with the RNA-seq results. The predicted *cis, trans*, and differentially expressed lncRNAs are listed in Table S7, Fig. 7.

3.8. Regulatory role of lncRNAs in the miRNA–mRNA network

Substantial evidence supports the coordinated regulation of mRNA

Fig. 5. Visualization of the network of stage-specific differentially expressed lncRNAs in four clusters. A–D The top 500 connections that link lncRNAs and neighboring mRNAs in cluster 3 (A), cluster 8 (B), cluster 11 (C), and cluster 13 (D). The white borders indicate mRNAs, and the purple borders indicate lncRNAs. The top thirty most important genes were predicted by CytoHubba, and a deeper red fill color indicates greater importance.

abundance by lncRNAs and miRNAs. Therefore, we identified differentially expressed miRNAs and screened for those that targered lncRNAs and mRNAs. TargetFinder was used to identify the differentially expressed miRNAs, and 5946 lncRNA endogenous target mimics (eTMs), associated with differentially expressed mRNAs were identified. Because lncRNAs modulate mRNA expression by competing for binding with miRNAs, we constructed a regulatory network to investigate the regulatory interactions among lncRNAs, miRNAs, and mRNAs. We divided the regulatory effects into two types: highly expressed miRNAs that corresponded to low-expression mRNAs and lncRNAs, and low-expression miRNAs that corresponded to highly expressed mRNAs and lncRNAs. In all six comparisons among the four developmental stages, miR408b was expressed in the downregulated miRNA network, and the upregulated miRNA network mainly contained members of the miR156x, miR827, miR396, miR399, and miR159 families. Based on trends in miRNA expression, we selected miR159, which contained two copies (miR159a and miR159c) in our results. miR159c exhibited low expression, and we therefore selected miR159a for further study. miR159 is a very ancient family that targets members of the GAMYB and GAMYB-like families at a highly conserved miR159 binding site. The families are named for their association with GA signaling and their MYB structural domain (Gubler et al., 1995). qRT-PCR results showed that

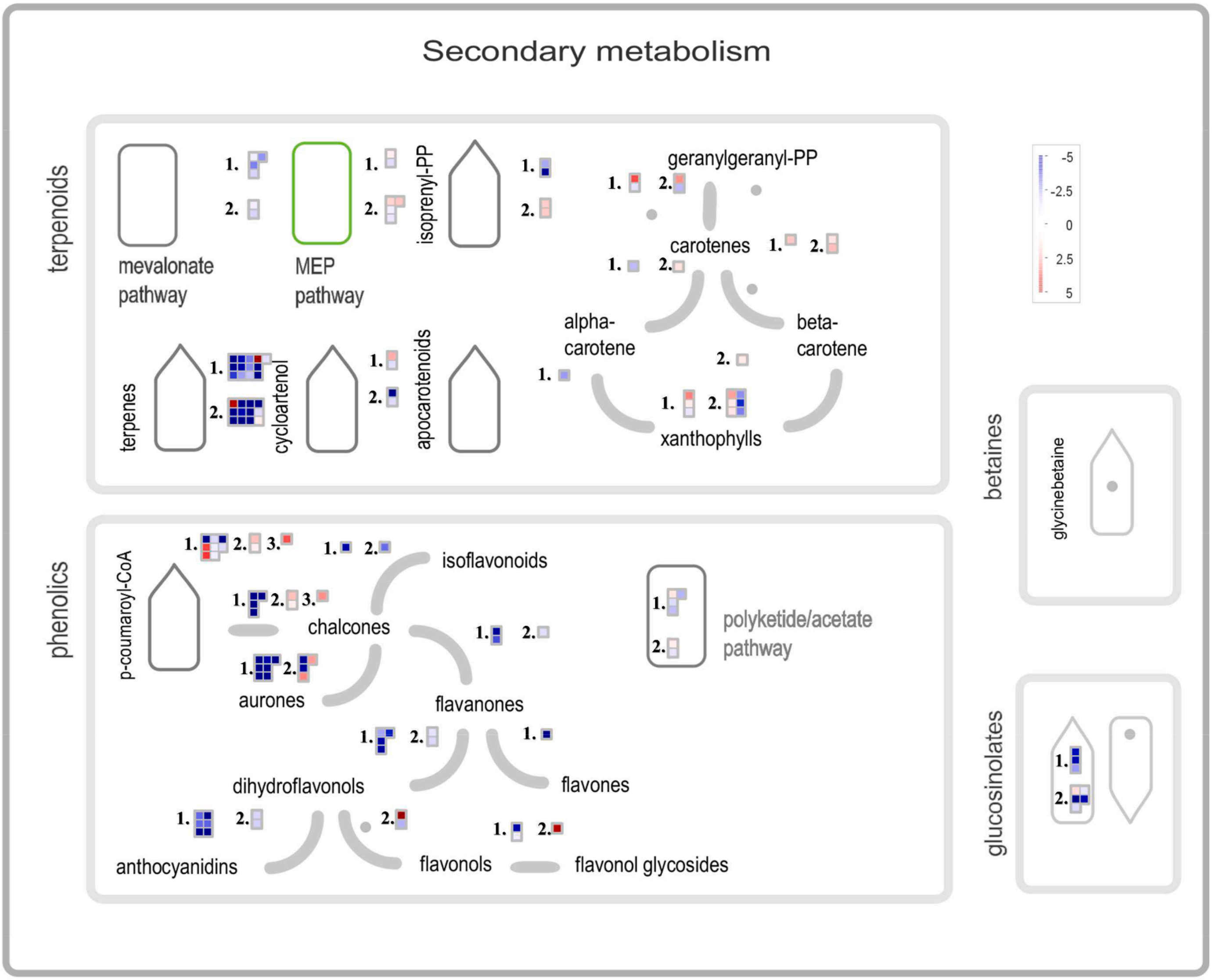

Fig. 6. Visualization of the co-expressed targets of differentially accumulated lncRNAs in secondary metabolism. Heatmap blocks show the expression of differentially expressed mRNAs targeted by differentially expressed lnRNAs in various secondary metabolic pathways. The numbers represent the different stage comparisons (1, PS2 vs. PS1; 2, PS3 vs. PS2; and 3, PS4 vs. PS3).

miR159a expression declined steadily during fruit development and remained relatively stable during PS3 and PS4. The expression of the GAMYB family target gene MYB33 (MD07G1085000) was higher during PS3 and PS4 than in PS1 and PS2 (Fig. 8A). We next constructed a miR159a-targeted lncRNA network in Cytoscape. lncRNAs targeted by miR159a were predicted, and positively (TCONS_00060619 and TCONS_00178112) and negatively correlated lncRNAs (TCONS_00065854) were considered to be eTMs or to have other potential functions in the network for expression verification (Fig. 8B). Our results suggest that there may be an association between these three lncRNAs and miR159-MYB33.

4. Discussion

Apple is one of the most popular, widely grown, and economically important crops in the world, and its fruit taste and quality are affected by a number of factors. lncRNAs (>200 nt) have important regulatory functions as non-coding RNAs, and recent research has focused on their important roles in plant flowering and DNA methylation (Chekanova, 2015). Previous work has revealed that lncRNAs participate in biological processes such as the iron deficiency response and IBA-induced adventitious root formation in apple (Meng et al., 2019; Sun et al., 2020). However, only a few lncRNAs involved in the process of anthocyanin accumulation have been reported during apple ripening (An et al., 2018; Ma et al., 2021; Yang et al., 2019). Here, we sequenced and annotated lncRNA and mRNA transcripts from apple fruit samples at 27, 84, 136, and 165 days after anthesis; these samples covered the juvenile, swelling, and trans-color fruit stages. The predicted lncRNAs had fewer exons, shorter ORFs, and shorter sequence lengths than mRNAs, consistent with previous studies (An et al., 2018; Zhang et al., 2018). Previous studies have focused on lncRNAs in the post-developmental stage of apple fruits (An et al., 2018), and therefore we compared our identified lncRNAs with those reported in previous work. Over 65% of the lncRNAs identified here were unique to our study, implying that they may function in early fruit developmental stages.

K-means clustering and Pearson correlation coefficients revealed stage-specific mRNA and lncRNA expression patterns and networks by identifying transcript clusters and co-expression networks. In the clustering results, Clusters 8 and 13 were highly expressed in the early stages of fruit development and were mainly associated with photosynthetic, auxin signaling and energy processes. They included four auxin response factor (ARF) genes (Chandler, 2016) and three squamosa promoter

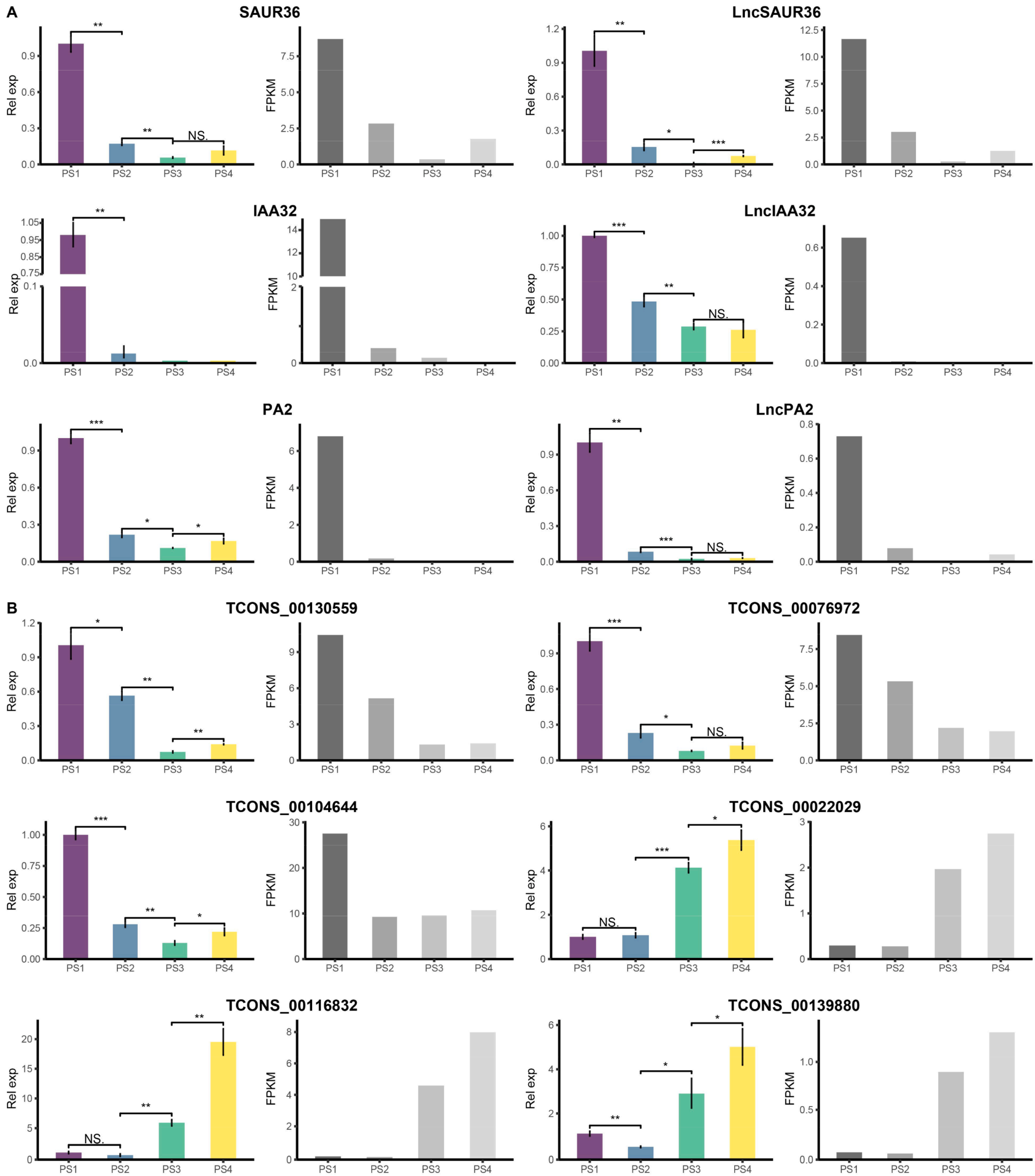

Fig. 7. Relative expression of differentially expressed genes and lncRNAs measured by qRT-PCR and RNA-seq. (A) High confidence, co-located lncRNA–mRNA pairs. (B) Three trans and three differentially expressed lncRNAs. The significance of comparisons was determined by Student's t-test.

binding protein-like (SPL) genes, which are known to regulate various plant developmental stages (Preston and Hileman, 2013). These clusters also contained genes encoding 23 light-harvesting complexes (LHCs) and light-regulated zinc finger protein 1 (LZF1), which participate in plant growth, development, light signaling (Datta et al., 2008; Zhao et al., 2020). By contrast, Cluster3, which was highly expressed during later fruit development, was mainly related to carbohydrates metabolism. It contained sucrose transporter 4 (SUT4), glucose-6-phosphate/phosphate translocator-related (CUE1), and phosphoglucose isomerase 1 (PGI1). The top 500 connections were used to construct co-expression networks for the four clusters (clusters 3, 8, 11, and 13). The key gene MD08G1099700, which has an unknown

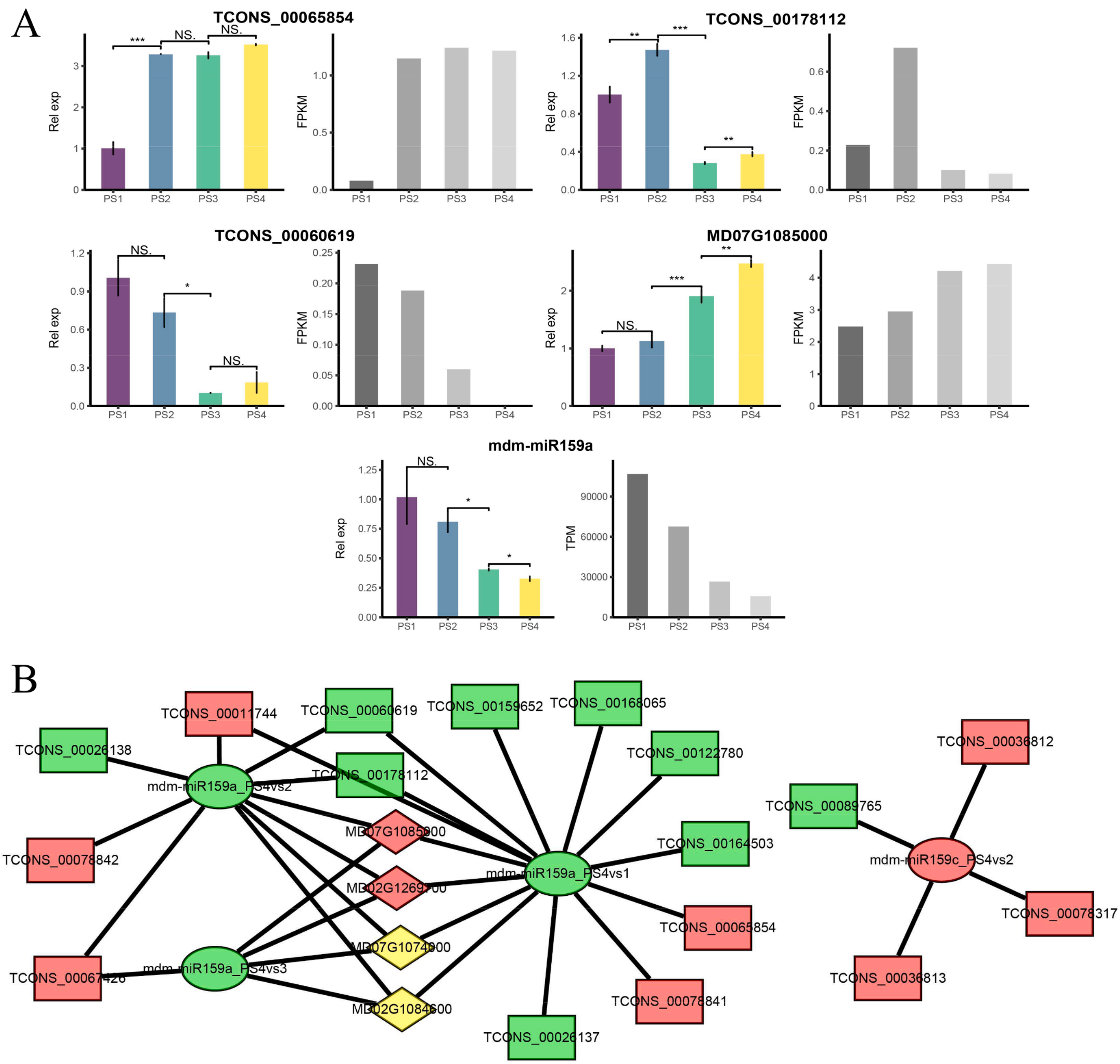

Fig. 8. Relative expression levels of lncRNAs, miR159, and mRNAs and their regulatory network. (A) The expression of three targeted lncRNAs, GAMYB, and miR159a. (B) The regulatory network of lncRNAs-miR159-mRNAs. The ellipses represent miRNAs, the rectangles represent lncRNAs, and the diamonds represent mRNAs. Green indicates downregulation between two comparison periods, red indicates upregulation, and yellow indicates both up- and downregulation.

function, was associated with three lncRNAs in the co-expression network of Cluster3 and may therefore be regulated by lncRNAs and play a role in ripening. In addition to the mRMA-lncRNA pairs with listed functions in the co-expression network results, some lncRNA–mRNA pairs contained auxin signaling genes (ARF3, SAUR-like) and stress-related genes (e.g., early-responsive to dehydration stress protein 4 [ERD4] and leucine-rich repeat [LRR]) in clusters 8, 11, 13. It is therefore possible that lncRNAs function in stress response during fruit ripening.

Interestingly, pathways related to the flavonoid metabolic process and phenylpropanoid metabolic process were detected in clusters 4 and 12, which had high expression in PS1 and PS4. This suggests that lncRNAs associated with trans-coloration are not only involved in PS4 but are also expressed in early fruit development. Genes in cluster 7, which were only highly expressed at PS4, were mainly enriched in catabolic process pathways. This suggests that lncRNAs linked to these pathways may play a role in the fruit pigmentation process. An overview of the clustering results revealed that most of the highly expressed lncRNAs (75.29%) were concentrated in PS1 and PS2, with a relatively small number of lncRNAs (16.8%) concentrated in PS3 and PS4. Likewise, 57.42% of the corresponding mRNAs were expressed during PS1 and PS2, and 25.27% were highly expressed during PS3 and PS4. This result suggests that lncRNAs have a greater regulatory role in the earlier stages of fruit development than in the later stages. Clustering analysis for individual periods of high expression showed that the numbers of lncRNAs continually decreased from PS1 to PS3 and then increased at PS4. This suggests that lncRNAs have some regulatory functions in early fruit development and at the trans-color stage consistent with previous reports on lncRNAs in apple (Ma et al., 2021; Yang et al., 2019). Notably, MD05G1311600 (ERF12-like protein) and MD07G1117200 (YABBY family protein) were found in the co-expression network of cluster 8, which was associated with flowering and early fruit development. Clustering analysis also showed decreased numbers of highly expressed lncRNAs in PS1 to PS3. lncRNAs may play important roles in developmental stages before PS1. Our experiments were limited to fruit development, but previous studies have shown that lncRNAs are also involved in flowering (Chekanova, 2015; He, 2012; Yamaguchi and Abe, 2012). Changes in the expression of lncRNAs from flowering to early fruit development would be an interesting direction for future research.

Numerous studies have shown that lncRNAs can regulate the expression of neighboring mRNAs (Deforges et al., 2019; Heo and Sung, 2011). We selected high-confidence, co-localized (within 10kbp), co-expressed lncRNAs for expression validation. Validation focused on MD13G1035200 (IAA32)-TCONS_00131720, MD02G1205700 (SAUR36)-TCONS_00025525, and MD03G1013600 (PA2)-TCONS_00032426, which have previously been linked to growth inhibition (Cao et al., 2019), leaf senescence (Hou et al., 2013), and phenylpropanoid biosynthesis. Several, trans lncRNAs and DE-lncRNAs that targeted metabolic processes were also selected for validation. The latter highly differentially expressed lncRNAs may contribute to the fruit development and carbohydrate accumulation. In addition to direct targeting, many lncRNAs play a regulatory role through the lncRNA–miRNA–mRNA network (Yang et al., 2019; Zhang et al., 2018). We identified endogenous target mimics (eTMs) and differentially expressed miRNAs, and we found three lncRNAs and two miR159 copies (miR159a and miR159c) that were predicted to target the GAMYB family of transcription factors. Although the miR159–GAMYB interaction is conserved, the conservation of its function is controversial. MiR159–SlGAMYB is required for fruit set in tomato, and high GAMYB levels induce parthenocarpy (da Silva et al., 2017). Interaction of miR159 with FaGAMYB in strawberries during receptacle development plays a role in the response to endogenous GA (Csukasi et al., 2012). Downregulation of FaGAMYB caused cessation of receptacle ripening, repressed color transformation, and reduced ABA and sucrose contents, demonstrating that FaGAMYB may be located upstream of the ABA and sucrose synthesis pathways (Vallarino et al., 2015). In Arabidopsis, the miR159–GAMYB pathway interacts with the miR319–TCP pathway in flower maturation. It forms a network with miR167 to promote ARF6/8 expression by repressing miR167 expression in sepals and petals, thereby affecting auxin signaling (Millar et al., 2019; Rubio-Somoza and Weigel, 2013). GAMYB may therefore play an important role in the ripening process of apples. The qRT-PCR verification results were consistent with the sequencing data and confirmed the reliability of the analysis. Our study provides a useful basis for subsequent studies of lncRNAs in apple fruits.

5. Conclusions

We performed strand-specific RNA-seq to identify lncRNAs expressed at four apple fruit developmental stages. The results showed that 12,570 mRNAs and 3571 lncRNAs were differentially expressed during the four stages. lncRNA–mRNA target relationships and their functions were predicted by k-means clustering and Pearson correlation analysis, co-expression network construction, and co-localization analysis. Endogenous target mimics (eTMs) of lncRNAs were predicted from miRNA data, and some high confidence transcripts were selected for qRT-PCR verification. These findings make an important contribution to research on lncRNAs during apple fruit development and ripening.

Data availability

The LncRNA and miRNA data have been deposited into the NCBI BioProject (BioProject, https://www.ncbi.nlm.nih.gov/bioproject/) under accession number PRJNA769977.

CRediT authorship contribution statement

Shicong Wang: Conceptualization, Methodology, Validation, Writing – original draft. **Meimiao Guo:** Validation, Data curation. **Kexin Huang:** Data curation, Visualization. **Qiaoyun Qi:** Formal analysis. **Wenjie Li:** Visualization. **Jinjiao Yan:** Writing – original draft. **Jieqiang He:** Methodology. **Qingmei Guan:** Supervision. **Fengwang Ma:** Supervision. **Jidi Xu:** Writing – review & editing, Investigation, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scienta.2022.110898.

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